MATH 1110
Calculus 1: Quiz 4
Given Time: 2 hrs.

Name: _____ Student ID: _____

Instruction. For each of the following problems, show all necessary steps in your work. Answers submitted without supporting work or with illegible handwriting may not be graded. Each question is worth the number of points indicated. You have **120 minutes** to complete the exam.

1. (10 points) Find the absolute maximum and absolute minimum values of

$$f(x) = x + \frac{1}{x} \qquad \text{on} \qquad [0.1, 4].$$

2. (10 points) Let

$$f(x) = x^3 - 3x + 2, \qquad \text{on} \qquad [-2, 2].$$

Verify that the function satisfies the hypotheses of the Mean Value Theorem on the given interval. Then find all numbers c that satisfy the conclusion of the Mean Value Theorem.

3. (10 points) Let

$$f(x) = |x - 3| \qquad \text{on} \qquad [1, 4].$$

Show that there is no value of c that satisfies

$$f'(c) = \frac{f(4) - f(1)}{4 - 1}.$$

Why does this not contradict the Mean Value Theorem?

4. (10 points) Show that the equation

$$x^3 + 15x - 4 = 0$$

has exactly on solution.

5. (**20 points**) Let $f(x) = x^3 + 3x^2 - 9x - 5$

a) Find the critical points of f

b) Find the interval(s) on which f is increasing and decreasing.

c) Use the first derivative test to determine the local max and local min of f.

d) Find the inflection point of f.

e) Find the interval(s) where f is concave up and concave down.

f) Sketch the graph of f.

6.(**10 points**) Evaluate the following limits.

$a)$ $\lim_{x\to 3}\left(\dfrac{x^2-9}{x^3-5x-12}\right)$ $b)$ $\lim_{x\to 0}\left(\dfrac{\sin x - x\cos x}{x - \sin x}\right)$

7.(**10 points**) A piece of wire of length 400 ft is bent into the shape of a rectangle. Which dimensions produce the rectangle of maximum area?

8.(**10 points**) Consider the following problem: a box with an open top is to be constructed from a square piece of cardboard, 1 meter wide, by cutting out a square from each of the four corners and bending up the sides. Find the largest volume that such a box can have.

9. (**10 points**) Use linear approximation to estimate $(8.3)^{1/3}$.

10. (**10 points**) Use Newton's method to approximate the zero of $f(x) = x^2 + 2x - 2$ in the interval [0,1].

a) Express x_{n+1} in terms of x_n

b) Find $x_1,\ x_2,\ x_3$, and x_4 starting at $x_0 = 0$. (Use four decimal places.)

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